

# Working Group VI

## Dwarf Galaxies in Cosmological Sims

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for the AGORA collaboration  
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# Dwarf galaxy halos cosmological runs

$$L_{box} = 5Mpc/h; V_{max} \sim 35; M_{vir} \sim 1E10M_{\odot}$$

- Reach high spatial and mass resolution (effective  $2048^3$ ):  
 $\epsilon \sim 45pc$  ( $0.5 \times \epsilon_{Power}$ ),  $M_g = 262M_{\odot}$ ,  $M_{dm} = 1.3E3M_{\odot}$
- Relevant to study many problems and for many working groups: Interstellar medium, Outflows, Characteristics of cosmological galaxies, etc...
- Nice start for cosmological hydrodynamical runs.

# Objectives of this Working Group

“Simulate and compare a  $10^{10} M_{\odot}$   
galactic halo across participating codes”

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⇓ Currently ⇓

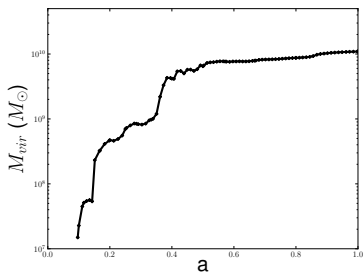
Make sure that all the pieces are put together  
so that the each group can start their  
dwarf galaxy hydrodynamical runs

Step by step approach

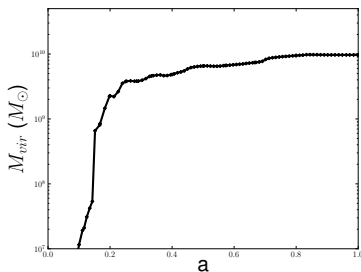
# 1. Two Halos selected following AGORA criteria ✓

| Halo | Type      | $M_{vir} (M_{\odot})$ | $R_{vir} (kpc)$ | $V_{max} (km/s)$ |
|------|-----------|-----------------------|-----------------|------------------|
| 001  | Merger    | 1.09E10               | 57.8            | 34.04            |
| 002  | Quiescent | 9.63E9                | 55.5            | 37.71            |

Merger

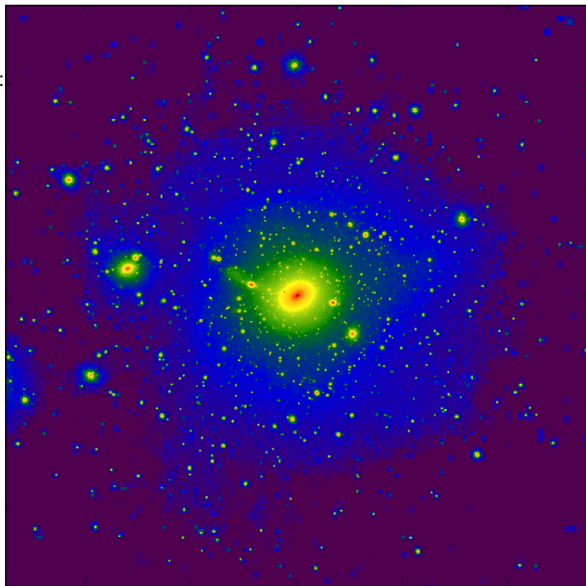


Quiescent



## 2. Create ICs and run test ✓

Lag. Vol. (cub.):  
 $\sim 1.2 \times 6 \times 6$   
( $Mpc^3/h^3$ )



Quiescent  
 $z = 0$

Gadget-2:  
200.000 hours

# Next step?

## Dark matter only runs

- + Good news all the tools are ready!!

## XSEDE Proposal next cycle

- + Deadline 15 October
- + Goals:
  - Test new implementations for ICs Lagrange Volumes
  - Dark matter only runs of  $10^{10} M_{\odot}$  halos
- + Any group/code interested in achieving these objectives is welcome to join.

# Codes & Volunteers in WG VI

|    | CODE         | VOLUNTEER         | DM |
|----|--------------|-------------------|----|
| 1  | ART-Chicago  | Sam Leitner       |    |
| 2  | ENZO         | Christine Simpson |    |
| 3  | GASOLINE     | Sijing Shen       |    |
| 4  | Nyx          | Wolfram Shmidt    |    |
| 5  | PKDGRAV2     | Mike Kuhlen       | ✓  |
| 6  | GADGET2-CFS  |                   |    |
| 7  | GADGET3      |                   | ✓  |
| 8  | GADGET3-SPHS |                   |    |
| 9  | GADGET3-AFS  |                   |    |
| 10 | ART-NMSU     |                   |    |
| 11 | Ramses       |                   |    |
| 12 | ...?         |                   |    |